

Mckenna Cisler

Software | Embedded | Controls | Robotics

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EDUCATION

B.S. Computer Science & B.A. Engineering, Brown University, Providence, RI GPA: 3.88 / 4.0 Sep 2016 — May 2020
Courses: Operating Systems, Algorithms, Machine Learning, Distributed Systems, Computer Networks, Collaborative Robotics, Digital Electronics Design, Design of Computing Systems, Signals & Systems, Advanced Digital Design, Control Systems, Communication Systems

TECHNICAL EXPERIENCE

Founding Firmware Engineer, Endurance Energy, Seattle, WA Dec 2025 — Present
— Building the firmware & controls stack for subsea geothermal power plants from scratch (in Rust).

Senior Software Engineer, SpaceX, Redmond, WA Sep 2020 — July 2025
— Rebuilt the radiation fault recovery system for Starlink satellites to automate 10+ engineer-hours of manual intervention per week. Reduced software development bottlenecks via a system that allowed engineers to self-serve fault recovery improvements while encoding best practices.
— Brought up a novel power system for a new satellite design from scratch, delivering on-time a safety-critical system that gated all vehicle-level testing. Aided hardware engineers in investigating and iterating on design flaws.
— Developed firmware for a novel BLDC motor control mode to enable very accurate antenna pointing and robust solar array actuation despite a suboptimal hardware design. Owned 3 independent actuator systems through bringup, first flight, reliability improvements, and handoff.
— Saved multiple individual \$500k satellites through rapid software fixes or diligent manual intervention. Became an authority on satellite-killer faults involving flight computers, software update, power systems, actuators, and ground comms.
— Fixed major reliability issues in backhaul system used for command & control to improve network performance and vehicle safety.
— Brought up next generation backhaul modem system. Leveraged strong ownership through the full design lifecycle to overcome design challenges and the project changing teams. Served users reliably within weeks of launch.
— Received 4 "Kickass Awards" for above accomplishments.

Graduate Flight Software Intern, NASA Jet Propulsion Laboratory, Pasadena, CA Jun 2020 — Aug 2020
— Developed flight software module in C for Psyche mission to a metal asteroid.

Avionics Intern, Boom Supersonic, Denver, CO May 2019 — Aug 2019
— Led clean-sheet design of mission-critical aircraft-tracking telemetry ground station, including requirement definition, critical design review, component integration, and C++ controller & user interface, for 5% of the cost of the commercial alternative.

Software Engineering Intern, The MITRE Corporation, Bedford, MA Jun 2018 — Aug 2018
— Built Android app to automate radio and power consumption testing. Enabled a 250-phone radio test without human participants.

Technical Lead / Flight Software Engineer, Brown Space Engineering, Brown University Sep 2016 — May 2020
— Led all technical projects as co-president of 75+ student team which designs and flies open-source CubeSats from scratch.
— Led preliminary design and writing of a NASA launch grant application for the team's next satellite.
— Worked on flight software (C-based RTOS), rad-safe bootloader, RF systems, ground software, and telemetry analysis for EQUiSat CubeSat. Operated on-orbit for >2 years.

Robotics Technician, Humanity-Centered Robotics Initiative, Brown University Sep 2016 — Dec 2018
— Led clean-sheet design and construction of hardware and software for "Walkerbot" elderly assistive robot (C++, ROS).
— Built analytics logging API and database for studying user interaction with an assistive toy (Node, C#, MongoDB).
— Prototyped ultrasonic localization system to help the elderly find household objects (Atmel MCU, Node).

PERSONAL PROJECTS

— Patent-pending actuated walker for the elderly to prevent falls; responsible for hardware bringup, software architecture, and control algorithm. 1st place in \$40K Analog Devices Sensor Fusion Challenge, Best Hardware Hack at Hack@Brown. www.jungyeop.com/args

SKILLS

Languages Strong Rust, C++, C, Python, Java; Proficient in Javascript, Go, Verilog, MATLAB
Tools Strong in FreeRTOS, ROS, Node, jQuery, Git; Proficient in React, OpenCV, Android; Experience with Simulink
Other Simulink, CAD (Inventor, OnShape, Blender), 3D Printing, Laser Cutting, Mill & Lathe
STM32, Atmel MCUs, FPGAs, Raspberry Pi, Arduino, and digital/analog circuits
Graphic design (Premier, Blender, GIMP, Photoshop, Inkscape)
Amateur radio operator (callsign KC1ICW), student pilot (51 flight hours), and FIRST Robotics (FRC) alum